Object Oriented Programming

(Practice MCQs)

1. Which of the following is not a feature of OOP in C++?

    a) Encapsulation

    b) Inheritance

    c) Polymorphism

    d) Compilation

2. What is encapsulation?

    a) Bundling data and methods that operate on the data into a single unit

    b) The ability to create a new class from an existing class

    c) The ability to redefine methods in derived classes

    d) None of the above

3. Which of the following access specifiers is not available in C++?

    a) Public

    b) Private

    c) Protected

    d) Friendly

4. Which keyword is used to define a base class in C++?

    a) base

    b) super

    c) class

    d) parent

5. Which type of inheritance is not supported directly by C++?

a) Single inheritance

b) Multiple inheritance

c) Multilevel inheritance

d) Hybrid inheritance

6. What is polymorphism in C++?

a) The ability of a function or operator to behave in different ways

b) The process of hiding data

c) The ability to create a new class from an existing class

d) None of the above

7. What is a virtual function in C++?

a) A function defined in a base class that can be overridden in a derived class

b) A function that exists in memory but is not used

c) A function that is called during object creation

d) None of the above

8. What is the output of the following code?

```
class Base {
public:
   void show() { cout << "Base" << endl; }
};

class Derived : public Base {
public:
   void show() { cout << "Derived" << endl; }
};

int main() {
```

```
    Base* b;
    Derived d;
    b = &d;
    b->show();
    return 0;
}
```

a) Base

b) Derived

c) Compilation error

d) Runtime error

9. What is the purpose of a constructor in C++?

a) To deallocate memory

b) To initialize objects

c) To create a new class

d) None of the above

10. Which of the following statements about destructors is true?

a) A class can have multiple destructors

b) Destructors are called manually by the programmer

c) Destructors are used to release resources

d) Destructors can be overloaded

11. What is the output of the following code?

```
class A {
public:
    A() { cout << "A"; }
    ~A() { cout << "~A"; }
};
```

```
int main() {
   A obj;
   return 0;
}
```

a) A

b) ~A

c) A~A

d) Compilation error

12. Which of the following is not a type of constructor in C++?

a) Default constructor

b) Parameterized constructor

c) Copy constructor

d) Virtual constructor

13. How is dynamic polymorphism achieved in C++?

a) Using overloaded functions

b) Using function overriding

c) Using function templates

d) Using default arguments

14. Which of the following can be declared as a friend in C++?

a) Function

b) Class

c) Another object

d) Both a and b

15. What is the output of the following code?

```
Class Base {
public:
   virtual void print() { cout << "Base"; }
};

class Derived : public Base {
public:
   void print() { cout << "Derived"; }
};

int main() {
   Base* b = new Derived();
   b->print();
   delete b;
   return 0;
}
```

a) Base

b) Derived

c) Compilation error

d) Runtime error

16. Which of the following is true about pure virtual functions?

a) They have no implementation in the base class

b) They must be implemented in the derived class

c) They are declared using the syntax = 0

d) All of the above

17. What is an abstract class in C++?

a) A class that cannot be instantiated

b) A class with at least one pure virtual function

c) A class with all its functions pure virtual

d) Both a and b

18. What is the use of the *this* pointer in C++?

a) To access the static members of the class

b) To differentiate between local and global variables

c) To access the object's members within the class methods

d) None of the above

19. What is the default access specifier for members of a class in C++?

a) Public

b) Private

c) Protected

d) None

20. Which of the following is correct about operator overloading in C++?

a) It allows defining new operators

b) It allows using operators with user-defined data types

c) It changes the syntax of the language

d) None of the above

21. What is the correct way to define a copy constructor?

```
Class A {
public:
```

```
    A(const A &obj) { /*...*/ }
};
```

a) A(const A obj) { /.../ }

b) A(A &obj) { /.../ }

c) A(A obj) { /.../ }

d) A(const A &obj) { /.../ }

22. Which of the following is a correct way to declare an array of objects in C++?

a) ClassName obj[5];

b) ClassName obj = new ClassName[5];

c) ClassName obj{5};

d) ClassName obj{};

23. Which of the following is true about inheritance in C++?

a) Derived class inherits private members of the base class

b) Derived class can access protected members of the base class

c) Derived class cannot override base class methods

d) None of the above

24. What does the *protected* access specifier mean?

a) Members are accessible only within the same class

b) Members are accessible within the same class and derived classes

c) Members are accessible within the same class and friend classes

d) Members are accessible from anywhere in the program

25. What is a virtual destructor in C++?

a) A destructor that does nothing

b) A destructor that can be called manually

c) A destructor that ensures derived class destructors are called

d) A destructor that can be overridden

26. What is the output of the following code?

```
class Base {
public:
   Base() { cout << "Base"; }
};

class Derived : public Base {
public:
   Derived() { cout << "Derived"; }
};

int main() {
   Derived obj;
   return 0;
}
```

a) Base

b) Derived

c) BaseDerived

d) DerivedBase

27. Which of the following is true about constructors and inheritance?

a) Base class constructor is called after derived class constructor

b) Derived class constructor is called after base class constructor

c) Constructors are not called in inheritance

d) Constructors are called in any order

28. How is operator overloading done in C++?

a) Using the *operator* keyword

b) Using function overloading

c) Using the *overload* keyword

d) Using inheritance

29. What does the *delete* operator do in C++?

a) Deletes an object from memory

b) Deletes a class

c) Deletes a function

d) Deletes an attribute

30. What is the output of the following code**?**

```
class A {
public:
   virtual void show() { cout << "A"; }
};

class B : public A {
public:
   void show() { cout << "B"; }
};

int main() {
   A* a = new B();
   a->show();
   return 0;
}
```

a) A

b) B

c) AB

d) Compilation error

Solutions:

1. d) Compilation
2. a) Bundling data and methods that operate on the data into a single unit
3. d) Friendly
4. c) class
5. d) Hybrid inheritance
6. a) The ability of a function or operator to behave in different ways
7. a) A function defined in a base class that can be overridden in a derived class
8. a) Base
9. b) To initialize objects
10. c) Destructors are used to release resources
11. c) A~A
12. d) Virtual constructor
13. b) Using function overriding
14. d) Both a and b
15. b) Derived
16. d) All of the above
17. d) Both a and b
18. c) To access the object's members within the class methods
19. b) Private
20. b) It allows using operators with user-defined data types
21. d) A(const A &obj) { /.../ }
22. a) ClassName obj[5];
23. b) Derived class can access protected members of the base class
24. b) Members are accessible within the same class and derived classes
25. c) A destructor that ensures derived class destructors are called
26. c) BaseDerived
27. b) Derived class constructor is called after base class constructor
28. a) Using the `operator` keyword
29. a) Deletes an object from memory
30. b) B